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Type-2 Fuzzy Logic Trajectory Tracking Control of Quadrotor VTOL Aircraft With Elliptic Membership Functions

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***Abstract*—Emerging applications of quadrotor vertical take-off and landing (VTOL) unmanned aerial vehicles (UAVs) in various fields have created a need for demanding controllers that are able to counter several challenges, inter alia, nonlinearity, underactuated dynamics, lack of modeling, and uncertainties in the working environment. This study compares and contrasts type-1 and type-2 fuzzy neural networks (T2FNNs) for the trajectory tracking problem of quadrotor VTOL aircraft in terms of their tracking accuracy and control efforts. A realistic trajectory consisting of both straight lines and curvatures for a surveillance operation with minimum snap property, which is feasible regarding input constraints of the quadrotor is generated to evaluate the proposed controllers. In order to imitate the outdoor noisy and time-varying working conditions, realistic uncertainties, such as wind and gust disturbances, are fed to the real-time experiment in the laboratory environment. Furthermore, a cost function based on the integral of the square of the sliding surface, which gives the optimal parameter update rules, is used to train the consequent part parameters of the T2FNN. Thanks to the learning capability of the proposed controllers, experimental results show that the efficiency and efficacy of the learning algorithms that the proposed T2FNN-based controller with the optimal tuning algorithm is 50% superior to a conventional proportional-derivative (PD) controller in terms of control accuracy but requires more controls effort. T2FNN structures are also shown to possess better noise reduction property as compared to their type-1 counterparts in the presence of unmodeled noise and disturbances.**

***Index Terms*—Quadrotor, unmanned aerial vehicles, UAV, aerial vehicles, fuzzy logic, fuzzy neural networks, tracking control, tyoe-2 fuzzy logic, elliptic membership functions**

I. Introduction

Quadrotor is a multi-rotor micro aerial vehicle that is lifted and propelled by four rotors. Due to their exceptional rotational agility, mechanical simplicity, relatively small size, vertical take off and landing (VTOL) ability and affordability, quadrotors have been the main interest of active research among the class of versatile flying robotic platforms. Moreover, due to its unmanned nature, its operation removes all costs and risks associated to onboard human pilots in both civilians and military fields, such as landscape mapping [1], agricultural surveying [2], aerial filming [3], surveillance [4], remote repair [5], search and rescue operations [6], delivery [7], multi-medium reconnaissance [8], and other fields where the environment is hazardous for human exposure [9]. Quadrotors' small size and low speed also enable them to maneuver through unknown difficult terrains with simultaneous localization and mapping [10]. Apart from individual vehicle missions, many researchers have also tried to exploit collective potential of quadrotor swarm to perform coordinated tasks such as constructing structures [11], crop monitoring [12], or even a spectacular light show [13].

Flight control system design is still a fundamental problem not only for quadrotors but also for aerial vehicles in general. Unlike typical helicopters, which rely on cyclic pitch to maneuver forward and sideways, a quadrotor has a constant fixed blade pitch on its propellers. Therefore, its motion can only be achieved by altering angular speed of each rotor. However, there are many challenges which are inherent to the quadrotors' dynamics. Thus, the controller design is not a straightforward task as its dynamics are highly nonlinear, underactuated, and having inter-couplings in its model. Furthermore, the case of having unknown working environment and both internal and external noise on the system demands a robust yet stable performance regardless of unmodeled external disturbances such as wind and drag. Amongst the aforementioned drawbacks, the most significant one is that the controller must operate around equilibrium point (usually hover), i.e. a small perturbation may drastically affect the output of the system.

Similar to other dynamic systems, there are two approaches for the control of quadrotors: those which need an exact mathematical model of the system (model-based controllers) and those which do not (model-free controllers). Proportional-integral-derivative (PID) and linear-quadratic regulators (LQRs) are the examples of the most widely used model-based controllers for their rule of thumb tuning simplicity. The aforementioned two controllers have been used to control an autonomous four-rotor micro helicopter in [14] which was one of the earliest real-time implementations. It is reported that both controllers provide mediocre results due to model approximations. In [15], another comparative study is given in which a PID, a classic LQR, and a PID tuned with an LQR loop controllers are compared, and it is reported that each controller offers singular characteristics that makes it hard to say which one is the best. In [16], a more complicated controller, model predictive controller (MPC), is applied to the control of quadrotors in uncertain environments where absolute-localization data are inadequate. The developed controller in [16] performs an accurate control in indoor flights by the utilization of inertial measurement unit (IMU)/sonar/optical flow data fusion system. However,

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in many quadrotor applications, a priori knowledge about the process or goals are just approximated. Therefore, these systems suffer from lack of modelling and both internal or external disturbances.

To obtain an accurate mathematical model of an aerial vehicle is a time consuming and tedious work. Parameter identification of an aerial vehicle requires specific experiments and each experiment needs to be repeated many times to get more accurate identification results [17]. In such cases, as an alternative to the model-based control, intelligent or model-free control may provide an easier solution as they do not need an accurate mathematical model of the system to be controlled. Without loss of generality, model-free controllers can be designed by either using the input-output relations of the system (e.g. artificial neural networks (ANNs) or expert knowledge (fuzzy logic controllers (FLCs))). ANNs are a family of supervised learning model that mimic the human central nervous system. They have been widely used in many applications including control engineering due to its ability to learn from input-output data. In [18], a sliding mode ANN control scheme consisting of two ANNs is proposed for quadrotor control in which it is reported that the amplification of measurement noise is eliminated without any performance degradation. On the other hand, fuzzy logic owes its exceptional scientific reputation to its unique ability to simultaneously deal with uncertainties in the system and use the expert knowledge as an input to the controller design. In [19], an FLC is designed and implemented successfully to control a quadrotor. Similarly in [20], a simple FLC with light computation volume is applied for quadrotor control. One weakness of FLCs is that they need to be frequently redesigned and tuned to cope with various uncertainties in the system. Manual tuning can be a very difficult and time consuming task to achieve good results. As a possible solution to this problem, the fusion of ANNs and FLCs are proposed which combines the ability of fuzzy systems to deal with ambiguousness with the learning capability of ANNs from input-output data sets. In the recent years, such a combination, namely fuzzy neural networks (FNNs) or Takagi-Sugeno-Kang (TSK) fuzzy models, has shown promising results as they integrate the advantages of FLCs and ANNs [21]–[25].

As an extension of type-1 fuzzy logic systems, their type-2 counterparts have also made tremendous impact on dealing with uncertainties over the last two decades as there are many sources of uncertainties associated with real-time aerial robot navigation such as changing working conditions, noisy data, imprecise actuation, unknown nonlinearity of the system, or difference in linguistic understanding of words used in antecedents and consequent part of rules. In contrast to ordinary type-1 fuzzy logic that has precise and crisp value of membership functions (MFs), type-2 fuzzy MFs are themselves fuzzy. The latter case results in that the antecedents and the consequent of the rules are totally uncertain thus giving the designers an extra degree of freedom to reach optimal performance. When the aerial robot parameters or working conditions vary, type-1 fuzzy sets, which were tuned for a specific condition, may not be appropriate anymore. Therefore, in such highly uncertain environments, type-2 FNNs (T2FNNs) appear to be a more promising method than their type-1 counterparts due to their flexibility by having infinite type-1 MFs combinations [26]. In [27], the noise reduction property of T2FNNs is proven to be superior to that of type-1 FNNs (T1FNNs) at higher noise level. However, at very low noise level, T1FNNs have comparable performance to T2FNNs, and hence is preferred because of their computational simplicity. Several researchers [28]–[31], have exploited type-2 fuzzy logic control architecture to autonomous mobile robots successfully.

In this paper, an interval T2FNN with elliptic type-2 fuzzy MFs working in parallel with a conventional proportional-derivative (PD) controller is proposed for the trajectory tracking problem of a quadrotor, and its performance is compared with its type-1 counterpart under wind and gust conditions. A cost function based on the integral of the square of the sliding surface, which gives the optimal parameter update rules, is used to train the consequent part parameters of the T2FNN. Thanks to learning capability of the proposed controllers, experimental results show the efficiency and efficacy of the learning algorithms that the proposed T2FNN-based controller with the optimal tuning algorithm is 50% superior to conventional PD controller in terms of control accuracy whilst requiring more control effort. To the best of our knowledge, this is the first time such optimal parameter update rules for the interval T2FNNs are tested in real-time by also considering various uncertainties. Additionally, this real-time implementation shows the efficiency and efficacy of T2FNNs for the first time to control highly nonlinear, underactuated and relatively fast dynamic aerial vehicles.

Paper organization: This paper is organized as follows: Section II starts with the kinematic and dynamical model of a quadrotor. In Section III, adaptive T1FNN and T2FNN with elliptic MFs are discussed. In section IV, we present the experiments and results. Lastly, some conclusions are drawn from this study in Section V.

II. Description of the Quadrotor Model

The schematic diagram of a quadrotor with its two sets of rotors (1, 3) and (2, 4) is illustrated in Fig. 1. Whereas rotor 1 and rotor 3 rotate clockwise, rotor 2 and rotor 4 rotate anticlockwise direction. This configuration is necessary to eliminate net torque around z axis such that zero reaction torques at hover can be achieved. By manipulating the angular speeds of each individual rotors, one can control six degrees-of-freedom (DOF) motion of the quadrotor through four inputs (roll, pitch, yaw, thrust). Specifically, in order to roll, a difference in angular speeds of rotor 2 and rotor 4 results in reaction torque along the x axis. Likewise, pitching is achieved by varying the angular speeds of the rotor 1 and rotor 3. Moreover, yawing motion is resulted from the balance torque of the counter-clockwise rotating rotors against the clockwise rotating rotors.

Unlike a conventional helicopter, a quadrotor has fixed pitch propeller hardware design. Therefore, the resulting thrust F_z of the rotors only points along the z axis of frame B which determines motion $(\ddot{X}, \ddot{Y}, \ddot{Z})$ in frame E.

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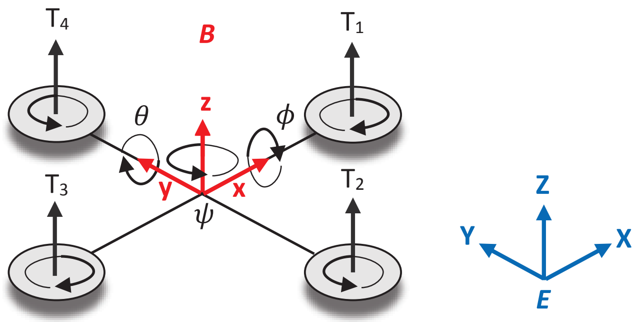

Fig. 1. Quadrotor model

A. Kinematics

The two reference frames are shown in Fig. 1: the Earth inertial frame E and the body-fixed frame B. In the body-fixed frame B, we assign (o,x,y,z) with o as the center of mass, x points towards the rotor 1, y points towards the rotor 4 and z points upwards. The linear position about the frame E is represented by (O,X,Y,Z) and the three Euler angles roll-pitch-yaw (ϕ,θ,ψ) are described as the orientation of the frame B with respect to the frame E. The linear velocities (U,V,W) and the body angular rates (p,q,r) are defined with respect to the B. The rotation matrix between the frame E and the frame B is given below (c:cos, s:sin):

$$R = \begin{bmatrix} c_\psi c_\theta & -s_\psi c_\phi + c_\psi s_\theta s_\phi & s_\psi s\phi + c_\psi s_\theta c_\phi \\ s_\psi c_\theta & c_\psi c_\phi + s_\psi s_\theta s_\phi & -c_\psi s_\phi + s_\psi s_\theta c_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix} \tag{1}$$

By using the rotation matrix above, any point on the airframe can be expressed in the frame E and the linear velocities of the quadrotor in the frame B can be projected onto to the frame E. For the angular rates, another transformation matrix is needed to transform $(\dot\phi,\dot\theta,\dot\psi)$ in the frame E to the frame B (p,q,r):

$$R_r = \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix} \tag{2}$$

B. Dynamics

In this paper, the quadrotor model is simplified as a rigid body with point mass with four rotors symmetrically distributed around the centre of mass. Applying the Newton's translational motion equation and Euler's rotational motion equation to vehicle dynamics, we can obtain six equations as follows:

$$\begin{aligned} \ddot X &= (\sin\psi\sin\phi + \cos\psi\sin\theta\cos\phi)\frac{F_z}{m} + \frac{D_x}{m} \\ \ddot Y &= (-\cos\psi\sin\phi + \sin\psi\sin\theta\cos\phi)\frac{F_z}{m} + \frac{D_y}{m} \\ \ddot Z &= -g + (\cos\theta\cos\phi)\frac{F_z}{m} + \frac{D_z}{m} \\ \ddot\phi &= \frac{(I_{yy}-I_{zz})\dot\theta\dot\psi - J_r\dot\theta\omega_r + \tau_x}{I_{xx}} \\ \ddot\theta &= \frac{(I_{zz}-I_{xx})\dot\phi\dot\psi + J_r\dot\phi\omega_r + \tau_y}{I_{yy}} \\ \ddot\psi &= \frac{(I_{xx}-I_{yy})\dot\phi\dot\theta + \tau_z}{I_{zz}} \end{aligned} \tag{3}$$

where m is the mass of the quadrotor and I_{xx}, I_{yy} and I_{zz} are the inertial components along the x, y and z directions in the frame B. The term F_z represents the vertical thrust along the z axis. The terms τ_x, τ_y and τ_z represent the torques associated to the thrust difference of each rotor pairs. The terms D_x, D_y and D_z are the drag forces for the velocities in X, Y and Z directions, J_r is the rotor inertia, and ω_r is the overall speed of the rotor as defined below:

$$\omega_r = -\omega_1 + \omega_2 - \omega_3 + \omega_4 \tag{4}$$

The relations between F_z, τ_x, τ_y, τ_z and each rotor's angular speeds can be obtained by analyzing the free body diagram of the model in Fig. 1. The terms ω_1, ω_2, ω_3 and ω_4 represent the angular speeds of the four rotors, b represents thrust factor, d represents drag factor, and l represents the length of each quadrotor's arm. The following matrix shows the relationships between F_z, τ_x, τ_y, τ_z and the angular velocities of the four propellers:

$$\begin{bmatrix} F_z \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} b & b & b & b \\ 0 & -lb & 0 & lb \\ -lb & 0 & lb & 0 \\ d & -d & d & -d \end{bmatrix} \cdot \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} \tag{5}$$

Since the aerodynamical effects are negligible and complicated to model in low velocities in which the quadrotor operates, they are ignored in the model in this study. The numerical values of the vehicle physical parameters used in the experiments are given in Table I.

III. Quadrotor Control

A. Control Architecture

A quadrotor is an underactuated force-controlled vehicle. Force actuation implies that rotational and translational commands are modelled as the second derivative of the attitude angle and position respectively. Due to the quadrotor's underactuation, only four DOFs out of its six DOFs (X,Y,Z,ϕ,θ,ψ) can be selected as control input variables. Therefore, cascaded

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TABLE I
DESCRIPTION OF THE QUADROTOR PARAMETERS [32]

Symbol	Description	Value
I_x	Moment of inertia about X axis	$0.007kgm^2$
I_y	Moment of inertia about Y axis	$0.007kgm^2$
I_z	Moment of inertia about Z axis	$0.012kgm^2$
J_r	Rotor moment of inertia	$6.5x10^{-5}kgm^2$
b	Thrust factor	$4.13x10^{-5}Ns^2$
d	Drag factor	$8.5x10^{-7}Nms^2$
l	Distance to the center of the quadrotor	$0.17m$
m	Mass of the quadrotor	$0.68kg$
g	Gravitational constant	$9.81m/s^2$

control system architecture must be implemented between two loops (outer and inner) to resolve this problem. In this paper, three positional coordinates and the corresponding yaw angle (X,Y,Z,ψ) are chosen as the states to be tracked.

The inner loop is responsible for the attitude control, and typically uses conventional PD controllers performed by the ground station as shown in Fig. 2. The outer loop is responsible for the position control of the quadrotor. The inputs to this control block are the position error measurements on the (X,Y,Z) axes and their derivatives with respect to the Earth frame obtained from the position sensors from VICON motion capture system. The outer loop then calculates the associated control inputs (u_x,u_y,u_z) required in each axis.

The three virtual control inputs, dimensionally similar to translational accelerations in (3), are introduced as follows:

$$\begin{bmatrix} u_x \\ \\ u_y \\ \\ u_z \end{bmatrix} = \begin{bmatrix} \dfrac{(\sin\psi\sin\phi+\cos\psi\sin\theta\cos\phi)F_z}{m} \\ \dfrac{(-\cos\psi\sin\phi+\sin\psi\sin\theta\cos\phi)F_z}{m} \\ -g+\dfrac{\cos\theta\cos\phi F_z}{m} \end{bmatrix} \tag{6}$$

By setting the reference yaw angle ψ_r equal to zero and by using the relationship between the virtual control signals and the attitude angles, the total force of the propellers F_z in the z direction and the reference attitude angles can be obtained as follows:

$$\begin{aligned} F_z &= m\sqrt{(u_z+g)^2+(u_x)^2+(u_y)^2} \\ \theta_r &= arctan\left(\frac{u_x}{u_z+g}\right) \\ \phi_r &= arcsin\left(-\frac{mu_y}{F_z}\right) \end{aligned} \tag{7}$$

The inner loop receives the attitude information to calculate the reference torques for each axis (τ_x,τ_y,τ_z). By using (5) as inverse dynamics, the obtained torques and thrust are then used to resolve the appropriate angular speeds for the four rotors.

B. Adaptive T2FNN Structure

The proposed controller scheme consists of an interval T2FNN, which is tuned by an optimal SMC approach working in parallel with a PD controller, is shown in Fig. 2. The PD controller is responsible for guaranteeing the global asymptotic stability in compact space and gain enough time for the initialization of the T2FNN to learn from the system dynamics. This control strategy is called feedback error learning and was firstly introduced by Kawato [33] for robot control. The PD control signal is as follows:

$$\tau_c = k_P e + k_D \dot{e} \tag{8}$$

where e is the feedback error, k_P and k_D are the proportional gains and derivative gains respectively.

The logic behind the feedback error learning scheme is as follows: To find the optimal values of a PD controller is similar to searching a point on a 2D surface. However, the proposed method works well with all the PD controller coefficients which makes the system stable for the initialization of the interval T2FNN. In other words, instead of finding the optimal point, we seek a circle around the optimal point. This approach is more practical in real-time control applications.

1) T2FNN Structure: In this study, A2-C0 fuzzy model is used in which the antecedents are type-2 fuzzy sets and the consequents are crisp numbers [34]. The ij^{th} rule of A2-C0 fuzzy systems with two inputs can be represented as follows:

$$IF\ x_1\ is\ \tilde{A}_{1i}\ and\ x_2\ is\ \tilde{A}_{j2},\ THEN\ \tau_f = f_{ij}\ i=1,...,I\ j=1,...,J \tag{9}$$

where x_1 and x_2 are the input variables, I and J are the number of MFs for x_1 and x_2 respectively, τ_f is the output variable, f_{ij} are the parameters of the consequent part of the fuzzy system, and $\tilde{A}_{1i}$, $\tilde{A}_{2j}$ are the type-2 fuzzy sets for the first and second input with their corresponding fuzzy MFs represented as $\tilde{\mu}_{1i}$, $\tilde{\mu}_{2j}$, respectively. The lower and upper MFs of $\tilde{\mu}_{1i}$ are $\underline{\mu}_{1i}$ and $\overline{\mu}_{1i}$. Similarly the lower and upper MFs of $\tilde{\mu_2}_j$ are $\underline{\mu}_{2j}$ and $\overline{\mu}_{2j}$, respectively.

The final output of the system can be represented as follows [34]:

$$\tau_f = q\frac{\sum_{j=1}^{J}\sum_{i=1}^{I}\underline{W}_{ij}f_{ij}}{\sum_{j=1}^{J}\sum_{i=1}^{I}\underline{W}_{ij}} + (1-q)\frac{\sum_{j=1}^{J}\sum_{i=1}^{I}\overline{W}_{ij}f_{ij}}{\sum_{j=1}^{J}\sum_{i=1}^{I}\overline{W}_{ij}} \tag{10}$$

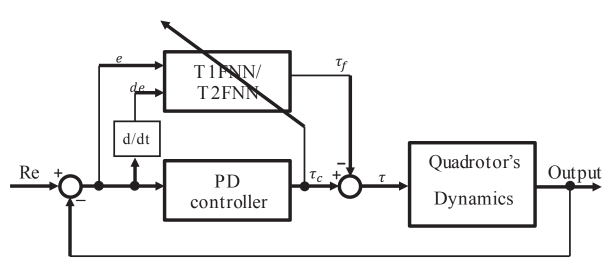

Fig. 2. Feedback error learning control strategy for the inner loop

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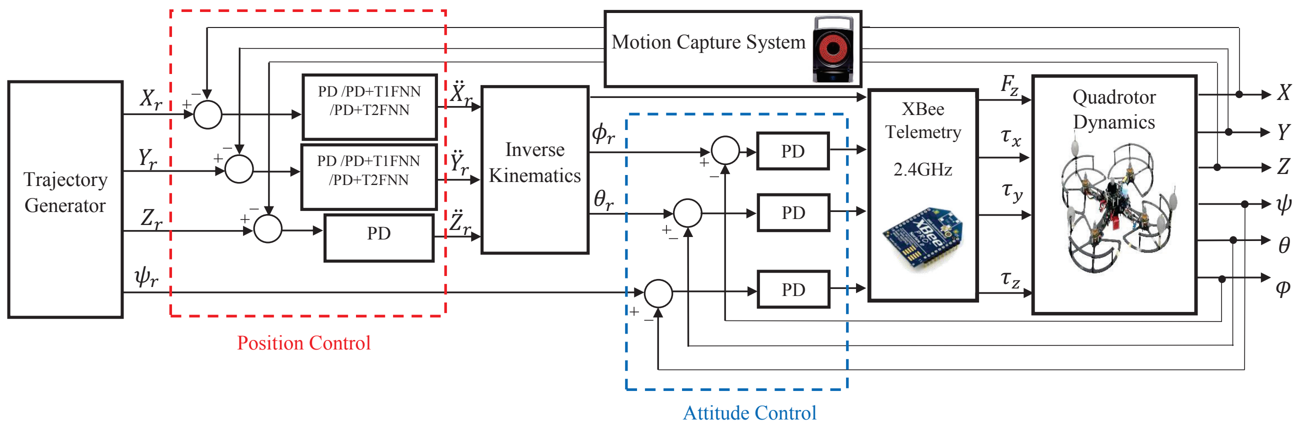

Fig. 3. The block diagram of the control scheme

where $\underline{W}_{ij} = \underline{\mu}_{1i}\underline{\mu}_{2j}$, $\overline{W}_{ij} = \overline{\mu}_{1i}\overline{\mu}_{2j}$, and the parameter q is the weighting gain that reflects the sharing of the contribution of the upper and lower MFs.

2) Elliptic Type-2 Fuzzy MFs: Elliptic MFs have certain values on the left and right end of the support, and it deals with uncertainty on the rest of the support. Inspired by the formulation of the triangular MFs, the lower and the upper MFs are defined with the following formulas:

$$\bar{\mu}(x) = \begin{cases} \left(1 - \left|\frac{x-c}{d}\right|^{a_1}\right)^{1/a_1} & \text{if } c-d < x < c+d \\ 0 & else \end{cases} \tag{11}$$

$$\underline{\mu}(x) = \begin{cases} \left(1 - \left|\frac{x-c}{d}\right|^{a_2}\right)^{1/a_2} & \text{if } c-d < x < c+d \\ 0 & else \end{cases} \tag{12}$$

where c and d are the center and the width of the MF, x is the input vector. The parameters a_1 and a_2 determine the width of the uncertainty of the proposed MF, and these parameters should be selected in the following form:

$$a_1 > 1 \tag{13}$$
$$0 < a_2 < 1$$

Figures 4(a), 4(b), 4(c) and 4(d) show the shapes of the proposed MF for $a_1 = a_2 = 1$, $a_1 = 1.2, a_2 = 0.8$, $a_1 = 1.4, a_2 = 0.6$ and $a_1 = 1.6, a_2 = 0.4$, respectively. As can be seen from Fig. 4(a), the shape of the proposed type-2 MF is changed to a type-1 triangular MF when its parameters are selected as $a_1 = a_2 = 1$.

The elliptic type-2 fuzzy MF has already been used for identification [35] and control [22], [36] purposes, and performed promising results.

In such a MF, the expert is very precise that a specific input around the center belongs to a specific fuzzy set. There is almost no uncertainty for the expert's viewpoint around the center. This is something what we expect from an expert that he/she is the person who chooses the centers of the MFs. What is more, as we are moving from the center towards the left or right end points of the support ($c+d$ or $c-d$), the expert is not certain anymore about his/her decision. It is subtle that after the half way over its support, the expert starts being more precise that a specific input does not belong to this fuzzy set.

C. Sliding Mode Control Theory-Based Learning Algorithm

The following SMC theory-based adaptation laws are designed to ensure the stability of the system in compact space. Let the zero dynamics of learning error coordinate $\tau_c(t)$ defined as a time-varying sliding surface s_c as follows:

$$s_c(\tau_f, \tau) = \tau_c(t) = \tau_f(t) + \tau(t) \tag{14}$$

Here, T2FNN structure is trained such way that the regulator will assist conventional parallel controller (in this case PD) so that desired performance can be achieved. The sliding surface for the nonlinear system is defined as:

$$s_p(e, \dot{e}) = \dot{e} + \lambda e \tag{15}$$

where λ is a positive variable that determines the reference trajectory of the error signal.

In this part, the adaptation theorem that summarizes the design process of the proposed method is presented.

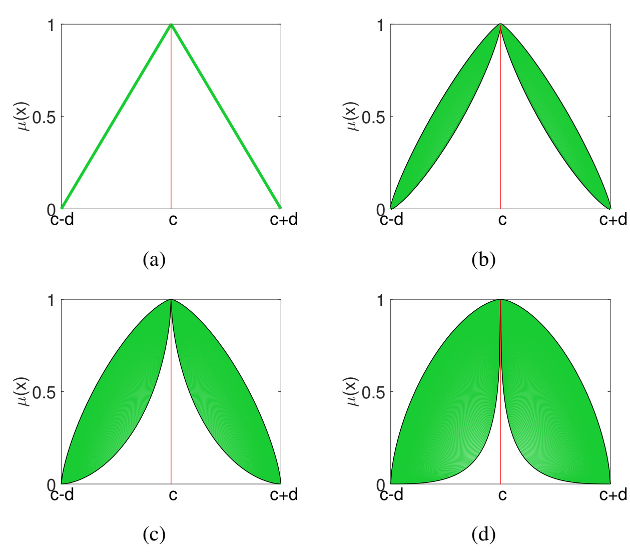

Fig. 4. The shape of the proposed type-2 MF for $a_1 = 1$ and $a_2 = 1$ (a) for $a_1 = 1.2$ and $a_2 = 0.8$ (b), for $a_1 = 1.4$ and $a_2 = 0.6$ (c), for $a_1 = 1.6$ and $a_2 = 0.4$ (d)

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Theorem 1: The adaptation laws for the consequent part parameters of T2FNN are chosen as:

$$\dot{F}(t) = -K_D P(t)\tilde{W}(t)s_p(t), \ F(0) = F_0 \in R^{n\times 1} \tag{16}$$

in which $P(t)$ is updated recursively as follows:

$$\dot{P}(t) = -P(t)\tilde{W}(t)\tilde{W}^T(t)P(t), \ P(0) = P_0 \in R^{n\times n} \tag{17}$$

where $\tilde{W}(t)$, $F(t)$ and $P(t)$ are defined as follows:

$$\begin{aligned}\tilde{W}(t) \quad = \quad & \Big[q\tilde{\overline{w}}_{11}(t) + (1-q)\underline{\tilde{w}}_{11}(t)\ldots \\ & q\tilde{\overline{w}}_{1J}(t) + (1-q)\underline{\tilde{w}}_{1J}(t)\ldots \\ & q\tilde{\overline{w}}_{ij}(t) + (1-q)\underline{\tilde{w}}_{ij}(t)\ldots \\ & q\tilde{\overline{w}}_{I1}(t) + (1-q)\underline{\tilde{w}}_{I1}(t)\ldots \\ & q\tilde{\overline{w}}_{IJ}(t) + (1-q)\underline{\tilde{w}}_{IJ}(t)\Big]^T\end{aligned}$$

$$F(t) = \Big[f_{11}(t)\ f_{12}(t)\ \ldots\ f_{1J}(t)\ \ldots\ f_{ij}(t)\ \ldots\ f_{I1}(t)\ \ldots f_{IJ}(t)\Big]^T$$

$$P(t) \quad = \quad \left[\int_0^t \tilde{W}(\tau)\tilde{W}^T(\tau)d\tau\right]^{-1}$$

Proof: Kindly see full proof in [36].

IV. Experimental Studies

Real-time flight tests for the trajectory tracking problem were conducted and evaluated in the Motion Capture Laboratory at Nanyang Technological University, Singapore. The indoor facility as shown in Fig. 5 was equipped with external optical motion capture system which provides real-time position and attitude measurement of a rigid body in a three dimensional space with an update rate of 100 Hz and accuracy level of 1 mm [32]. Retro-reflective markers were affixed to the quadrotor in a unique pattern to provide visual feedback of its predefined body frame for recognition. Three markers would be sufficient to compute all six DOFs of the vehicle. However, for redundancy reasons and to compensate individual shifts, five markers are used as shown in Fig. 6. Image data from 20 MX-VICON infrared cameras were sent to the ground computer via Ethernet connection. Then, VICON tracking software processes these flight data into six DOFs information as an input to our control system in MATLAB/Simulink. This platform provides accurate and high sample rates for state estimation. Therefore, the precision of this localization and pose recognition setup is sufficient and eliminates the need of data fusing with other on-board navigation sensors, such as LIDAR or GPS.

Motor speed pulses to electronic speed controllers (ESCs) are provided by Pixhawk embedded flight controller implemented onboard of the quadrotor. Position and attitude control are performed by proposed controllers on ground computer using MATLAB/Simulink platform which runs at the rate of 50 Hz. Communication with the ground computer is realized via XBee telemetry module over a serial port which transmits desired thrust F_z and 3 angular torques (τ_x, τ_y, τ_z) input values.

In order to demonstrate the noise reduction property of the proposed controller, we consider adding external disturbances with 2 identical industrial fans with a flowrate of 2.5 m^3/s and 3 m^3/s for low speed and high speed mode respectively. With these large flowrates, a large portion of the trajectory can be covered. During the simulation, maximum translational velocity is kept to be 2 m/s and maximum acceleration of 0.7 m/s^2. Each controller types and wind speeds scenario was retested three times to avoid random errors.

A. Trajectory Generation

Throughout the real-time experiments, a trajectory is defined with the following conditions:

1) The proposed 3D path is a time-based trajectory.
2) The trajectory consists of different path types such as curves, hover and straight lines.
3) The trajectory is feasible by means of the states and input constraints of the quadrotor.
4) The trajectory is optimal trajectory which minimizes snap.

In the proposed optimal trajectory, the autonomous navigation of the vehicle combines several manoeuvres such as VTOL, climbing and ascending curves as well as straight lines. The quadrotor first hovers up to a 1 m height followed with a straight diagonal climb to 2 m height. After climbing, the

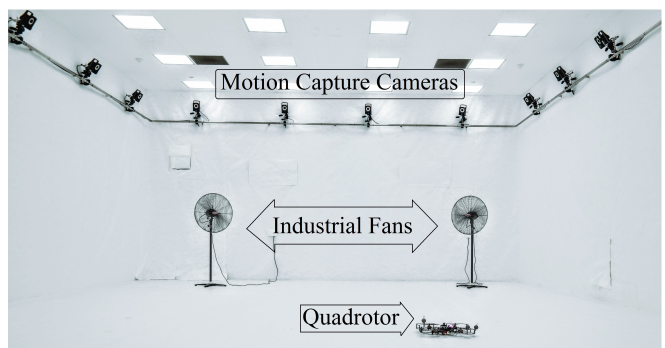

Fig. 5. Motion Capture Lab setup with infrared cameras, industrial fans, and aerial vehicle

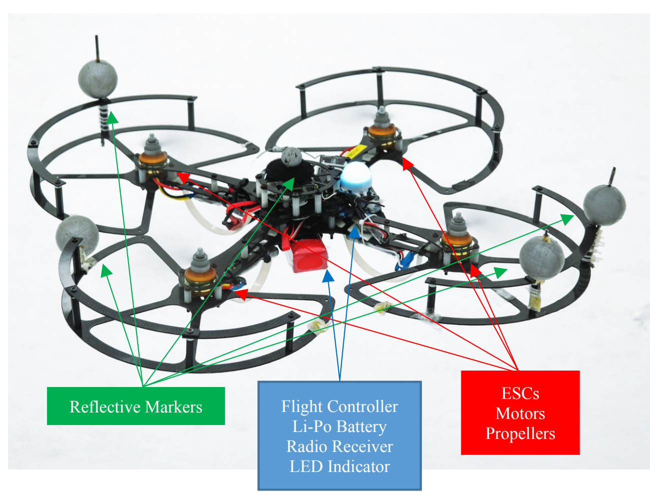

Fig. 6. The quadrotor used throughout the experiments in the Motion Capture Laboratory at Nanyang Technological University, Singapore

quadrotor makes a level circle and straight line. The quadrotor then descends, starts turning at opposite direction and level curve before coming back to its initial position. Here, the purpose of such '8' shaped trajectory is to test the robustness of each controller in a full flight sequence manner which is commonly used in real-time surveillance operations. The generated waypoints are ensured to be inside the quadrotor's dynamic, structural and aerodynamic safety operating limits. In addition, the control input saturation is implemented to restrict the desired performance under the quadrotor's dynamic constraints.

Without loss of generality, consider a navigation problem of reaching m waypoints at specified time at position $n = [n_0, n_1, ...n_m]$. The common practice is to connect these waypoints with straight lines. However, this method always results in inefficient trajectories because the vehicle has to encounter several sharp turns. As the quadrotor is a fourth order system, a trajectory that minimizes snap (fourth derivative of position) is created for smooth transitions between the waypoints [37]. Let p denotes position of the vehicle with respect to ground frame $p = [X, Y, Z]$. By Euler-Lagrange equation:

$$p*(t) = argmin_{p(t)} \int_0^T L(p, \dot{p}, \ddot{p}, \dddot{p}, \ddddot{p}, t)dt \tag{18}$$

$$L = \ddddot{p}^2 \tag{19}$$

The resulting equation between two consecutive waypoints is a seventh order polynomial in time.

$$p_i(t) = \alpha_{0i} + \alpha_{1i}t + \alpha_{2i}t^2 + ... + \alpha_{6i}t^6 + \alpha_{7i}t^7 \tag{20}$$

Define piecewise continuous trajectory function as,

$$p(t) = \begin{cases} p_1, & t_0 \le t < t_1 \\ p_2, & t_1 \le t < t_2 \\ .... & \\ p_m, & t_{m-1} \le t < t_m \end{cases} \quad T = [t_0, t_1...t_m]^T p = [p_1, p_2...p_m]^T \tag{21}$$

By specifying the initial condition values, intermediate condition as $p_i(t_i) = p_{i+1}(t_i) = n_i$, final condition values, and all the position derivatives as $d^k p_i/dt^k(t_i) = d^k p_{i+1}/dt^k(t_i)|_{t=T}$ in order to be smooth, it is sufficient to solve for all required coefficients for the piecewise trajectory. After computing the trajectory, desired velocity and acceleration can be obtained by differentiating the trajectory function. For the feasibility check of the generated trajectory, inverses of the dynamics are applied to ensure required motor speed is within the limits.

B. Experimental Results

The performances of the PD controller only and the T2FNN working in parallel with a conventional PD controller for the trajectory tracking without the presence of wind are presented in Figs. 7-8. As can be seen from these figures, the PD controller has a significant steady state error that persists throughout the trajectory. On the other hand, due to its optimal sliding mode learning capability, the T2FNN compensator is able to reduce this steady state error and hence trace closer to the reference trajectory.

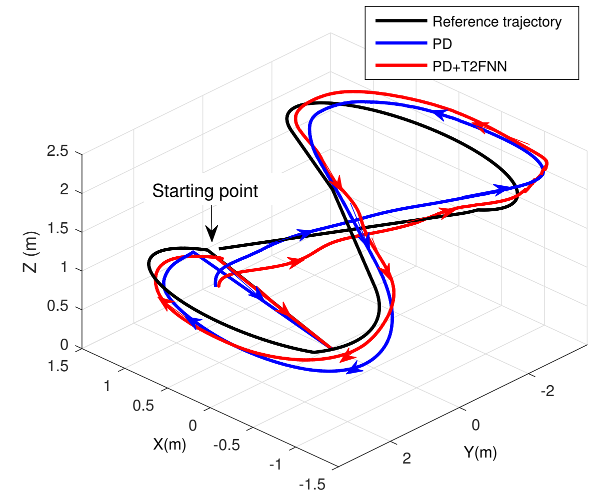

Fig. 7. Trajectory tracking performance of the different controllers

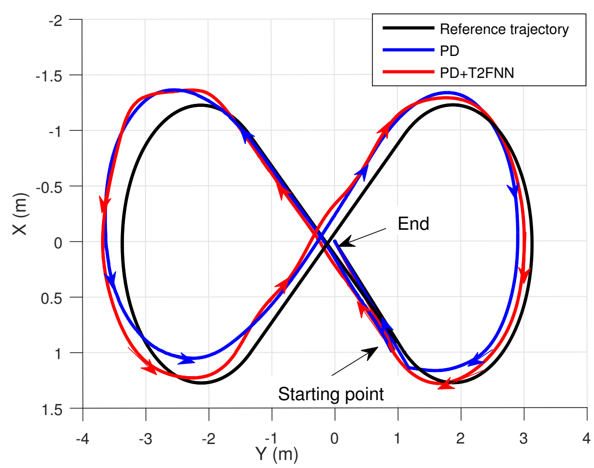

Fig. 8. Trajectory tracking performance of the different controllers in X-Y plane

Figure 9 shows the Euclidean error values of the PD controller alone and FNN working along with PD. It can be noticed that the performance of the T2FNN with the PD is superior throughout the trajectory. The T2FNN is able to reduce significant amount of error from peak of around 1.3 m to just 0.8 m. Of course, we can tune the PD more aggressively for better performance in each specific scenario. However, it is not practical for real time applications in which trajectories needed to be online generated, and uncertainties of the environment are present. Another serious problem with an aggressive controller is that when the controller is tuned w.r.t straight lines, it gives a bigger steady state error for a curve line trajectory. In addition, the controller that is tuned w.r.t curve lines gives an oscillatory response for straight lines.

In cases where wind disturbances are present, we divide the experiments into two groups: low wind and high wind scenarios. The performance of each controller in terms of position errors is presented in Figs. 10-11. The T2FNNs implemented in the experiments are of elliptical MF parameters

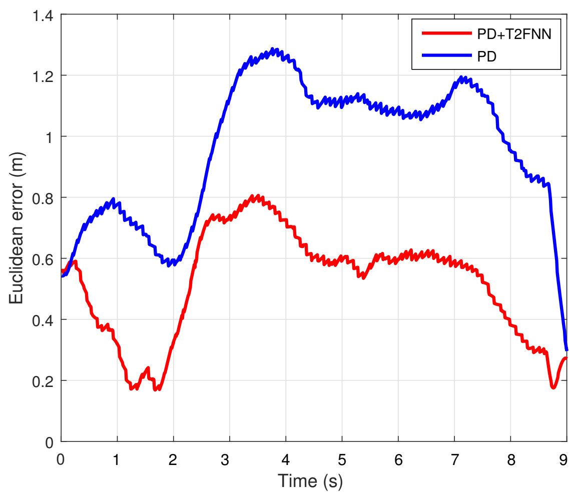

Fig. 9. Euclidean error of the different controllers in absence of wind

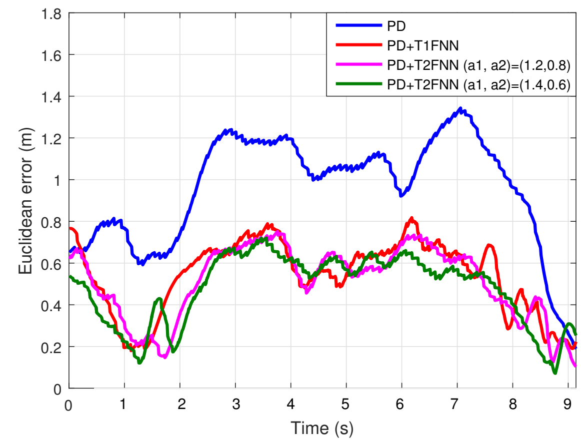

Fig. 11. Euclidean error of different controllers' performance in presence of high wind

(1.2,0.8) and (1.4, 0.6). In both cases, T1FNN and T2FNN structures have a notably less error as compared to the PD controller working alone. Interestingly, in an environment with wind gusts, T2FNN tracking results are slightly better than T1FNN due to its noise handling capability. T2FNN control structures are of less error peaks as well demonstrating it to be more stable even if there exist unmodeled dynamics and uncertainties in real environments. These unmodeled dynamics may include nonlinearities such as wind, propeller flexibility, or aerodynamic drag. Therefore, the intelligent structure consisting of T2FNN with a conventional PD controller would be preferable in a real-time application.

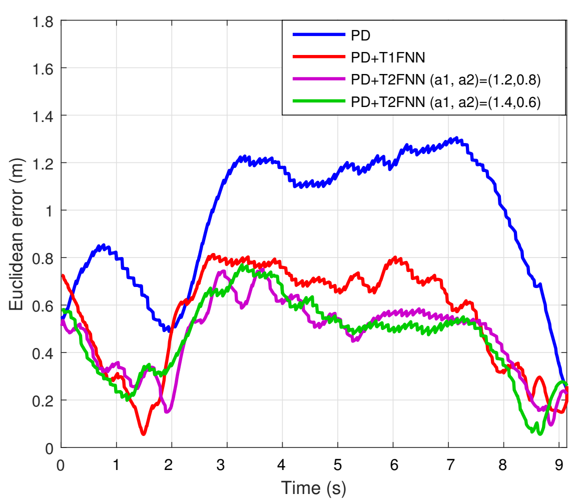

Fig. 10. Euclidean error of different controllers' performance in presence of low wind

The control signals for the cases of having the PD controller working alone and the T2FNN in parallel with the PD controller can be seen in Figs. 12-13. It is observed that the intelligent compensator, T2FNN, is able to take over the responsibility of controlling the system, and thus it gradually replaces the PD controller. This results in a tendency to a zero output from the PD controller. In other words, after a finite time, the PD controller does not produce any outputs, but only the T2FNN manipulates the system. The output of the PD controller becomes nonzero when the trajectory sequence shifts from one trend to another. After such a transient time, the T2FNN learns the system dynamics again, and eventually replaces the PD controller.

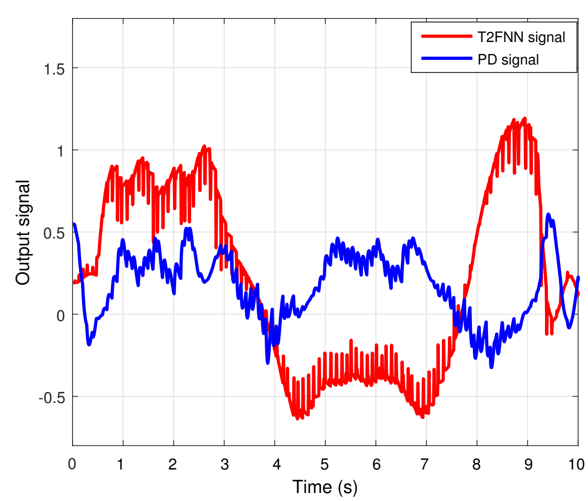

Fig. 12. Comparison of PD and T2FNN output signals on X axis

The root mean squared error (RMSE) values and the control efforts for each axis are listed in Table II. Three data values for each scenario were collected, and their mean is also displayed on the right cell of each experiment. It is observed that in the absence of wind disturbance, T2FNN decreases the PD controller RMSE error by about 43%. In the presence of wind disturbances, T2FNN structures again have an advantage over the conventional PD controllers. Moreover, the T2FNN controllers can further suppress noise effects by 2% to 7% as compared to their type-1 counterparts. Here, it is also noticed that the larger the noise presents, it is favorable to widen the

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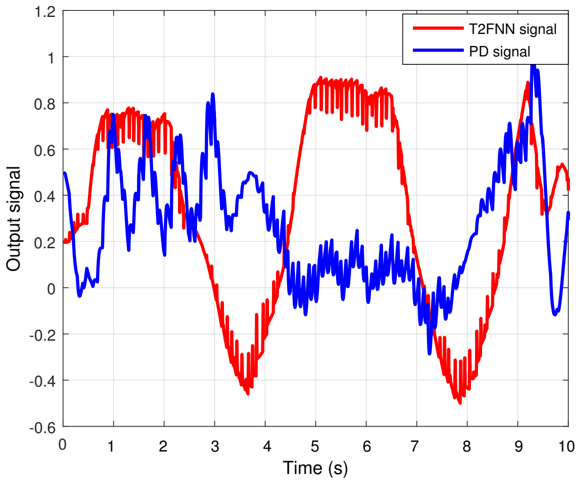

Fig. 13. Comparison of PD and T2FNN output signals on Y axis

range of type-2 fuzzy elliptic MFs.

As can be seen from Table II, it is proven that the PD controllers are the most efficient one as they require the least control effort. On the other hand, more precise maneuvers of T2FNN working along with the PD controller requires higher control effort. In practice, the proposed T2FNN structure does not require computationally expensive calculations such as large matrix inverses which exhaust resources and induce delays.

TABLE II
RMSE AND CONTROL EFFORT

Controller Types	RMSE		Control Effort X Axis		Control Effort Y Axis	
No Wind						
PD	0.978		0.150		0.431	
	0.957	0.972	0.233	**0.199**	0.284	**0.403**
	0.981		0.216		0.493	
PD+T2FNN	0.544		0.207		1.496	
	0.552	**0.547**	0.533	0.344	1.273	1.372
	0.544		0.291		1.346	
Low Wind						
PD	1.011		0.264		0.634	
	0.942	0.970	0.261	**0.238**	0.538	**0.547**
	0.955		0.189		0.468	
PD + T1FNN	0.598		0.900		3.543	
	0.539	0.544	0.895	0.726	1.398	1.841
	0.494		0.383		0.582	
PD + T2FNN (a_1,a_2)**=(1.2,0.8)**	0.509		0.478		1.430	
	0.516	0.507	0.588	0.445	1.564	1.297
	0.497		0.267		0.898	
PD + T2FNN (a_1,a_2)**=(1.4,0.6)**	0.482		0.361		0.925	
	0.538	**0.505**	0.647	0.422	1.416	1.091
	0.496		0.259		0.932	
High Wind						
PD	0.991		0.538		0.446	
	0.934	0.954	0.391	0.424	0.688	**0.564**
	0.938		0.343		0.557	
PD + T1FNN	0.551		0.708		1.367	
	0.549	0.555	0.487	0.702	0.642	1.178
	0.566		0.910		1.524	
PD + T2FNN (a_1,a_2)**=(1.2,0.8)**	0.552		0.383		0.588	
	0.539	0.547	0.301	**0.345**	1.421	0.977
	0.550		0.352		0.923	
PD + T2FNN (a_1,a_2)**=(1.4,0.6)**	0.505		0.385		1.679	
	0.547	**0.526**	0.564	0.574	1.154	1.728
	0.525		0.774		2.350	

V. CONCLUSION

This paper studies the trajectory tracking problem of a nonlinear quadrotor aerial vehicle in the presence of wind gust conditions. We propose T1FNN and T2FNN with elliptical MFs tuned by an optimal SMC theory-based algorithm to overcome the issue of lack of modeling and time varying working conditions. A realistic and optimal trajectory tracking task, which respects dynamic and input constraints of the quadrotor, is generated. Extensive real-time quadrotor flight tests are conducted to compare and verify the performance of the proposed controllers with regard to a conventional PD controller. Experimental results show that the T1FNN and T2FNN working in parallel with a PD controller have significantly lower steady state error as compared to a sole PD controller. However, PD controllers are most efficient in terms of their control effort. The proposed T2FNN also shows better noise rejection property by implementing elliptic MFs as compared to T1FNN in circumstances where high wind gusts are present. As a future study, the width of the elliptic MFs can be tuned in an adaptive manner throughout the real-time operation.

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